Homework 8: Normalization

Question 1

Functional Dependencies:

```
Invoice No → Sale Date
Product No → Prod Desc, Vend Code, Vend Name, Prod Price
Vend Code → Vend Name
{Invoice No, Product No} → Qty Sold
```

Partial Dependencies (violate 2NF):

• Invoice No → Sale Date (depends on only half of the composite key)

• Product No → Prod Desc, Vend Code, Vend Name, Prod Price (depends on only half of the composite key)

Transitive Dependency (violates 3NF):

• Product No → Vend Code → Vend Name

Step 1 — Decompose to 2NF (remove partial dependencies):

INVOICE (<u>Invoice No</u>, Sale Date)
PRODUCT (<u>Product No</u>, Prod Desc, Vend Code, Vend Name, Prod Price)
INVOICE_LINE (<u>Invoice No</u>, <u>Product No</u>, Qty Sold)

Step 2 — Decompose to 3NF (remove transitive dependency):

PRODUCT contains: Product No → Vend Code → Vend Name. Remove Vend Name and create a VENDOR table.

Final Tables (3NF):

INVOICE (<u>Invoice No</u>, Sale Date)
PRODUCT (<u>Product No</u>, Prod Desc, Vend Code, Prod Price)
VENDOR (<u>Vend Code</u>, Vend Name)
INVOICE_LINE (<u>Invoice No</u>, <u>Product No</u>, Qty Sold)

Question 2

Primary key: StuID

Functional Dependencies:

```
StuID → StuName, StuMajor, DeptID, StuGPA, StuHrs, AdvName
DeptID → DeptName, DeptPh, College
AdvName → AdvOffice, AdvBldg, AdvPh
StuHrs → StuClass
```

Transitive Dependencies (violate 3NF):

• StuID → DeptID → DeptName, DeptPh, College

• StuID → AdvName → AdvOffice, AdvBldg, AdvPh

• StuID → StuHrs → StuClass

Final Tables (3NF):

```
STUDENT (StuID, StuName, StuMajor, DeptID, StuGPA, StuHrs, AdvName)
DEPARTMENT (DeptID, DeptName, DeptPh, College)
ADVISOR (AdvName, AdvOffice, AdvBldg, AdvPh)
CLASS (StuHrs, StuClass)
```

Question 3

The original table is not in 1NF because VISIT DATE and PROCEDURE contain multiple values per pet.

Step 1 — Normalize to 1NF:

Flatten the table so each row has one visit date and one procedure. Composite primary key: {PET ID, VISIT DATE}.

```
PET_VISITS (PET ID, PET NAME, PET TYPE, PET AGE, OWNER, VISIT DATE, PROCEDURE)
```

Step 2 — Normalize to 2NF:

PET NAME, PET TYPE, PET AGE, and OWNER depend only on PET ID, not on the full composite key. Remove these partial dependencies.

```
PET (PET ID, PET NAME, PET TYPE, PET AGE, OWNER)
VISIT (PET ID, VISIT DATE, PROCEDURE)
```

Step 3 — Normalize to 3NF:

No transitive dependencies exist in either table. The 2NF result is already in 3NF.

Final Tables (3NF):

```
PET (PET ID, PET NAME, PET TYPE, PET AGE, OWNER)
VISIT (PET ID, VISIT DATE, PROCEDURE)
```

Question 4

Primary key: {PartNumber, Supplier}

Functional Dependencies:

```
PartNumber → Description
Supplier → SupplierAddress
{PartNumber, Supplier} → Price
```

BCNF Violations:

• PartNumber → Description: PartNumber is not a superkey → violates BCNF

• Supplier → SupplierAddress: Supplier is not a superkey → violates BCNF

Decompose to BCNF:

Extract each violating FD into its own table. The remaining table keeps the original composite key and Price.

Final Tables (BCNF):

```
PART (PartNumber, Description)
SUPPLIER (Supplier, SupplierAddress)
PART_SUPPLIER (PartNumber, Supplier, Price)
```